

Fourier Series examples solved

AGH Electronics and Telecommunications

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Expanding 2π - Periodic Function into AGH a trigonometric Fourier series

If a 2π - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -\pi, \pi \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)]$$

where coefficients are given by the formulae

$$a_0 = \frac{1}{\pi} \int_{2\pi} f(x) dx \quad (1)$$

$$a_n = \frac{1}{\pi} \int_{2\pi} f(x) \cos nx dx, \quad n \geq 1 \quad (2)$$

$$b_n = \frac{1}{\pi} \int_{2\pi} f(x) \sin nx dx, \quad n \geq 1 \quad (3)$$

where integrations are over any interval in x of the length 2π .



Expanding $2l$ - Periodic Function into AGH a trigonometric Fourier series

If a $2l$ - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -l, l \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{n\pi x}{l} + b_n \sin \frac{n\pi x}{l},$$

where coefficients are given by the formulae

$$a_0 = \frac{1}{l} \int_{-l}^l f(x) dx \quad (4)$$

$$a_n = \frac{1}{l} \int_{-l}^l f(x) \cos \frac{n\pi x}{l} dx, \quad n \geq 1 \quad (5)$$

$$b_n = \frac{1}{l} \int_{-l}^l f(x) \sin \frac{n\pi x}{l} dx, \quad n \geq 1. \quad (6)$$



Expanding l - Periodic Function into AGH a trigonometric Fourier series

If a l - periodic function f satisfies the Dirichlet Conditions then it can be expanded at any point $x \in \langle -\frac{l}{2}, \frac{l}{2} \rangle$ to its Fourier series ie.

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{+\infty} a_n \cos \frac{2\pi nx}{l} + b_n \sin \frac{2\pi nx}{l},$$

where coefficients are given by the formulae

$$a_0 = \frac{2}{l} \int_{-\frac{l}{2}}^{\frac{l}{2}} f(x) dx \quad (7)$$

$$a_n = \frac{2}{l} \int_{-\frac{l}{2}}^{\frac{l}{2}} f(x) \cos \frac{2\pi nx}{l} dx, \quad n \geq 1 \quad (8)$$

$$b_n = \frac{2}{l} \int_{-\frac{l}{2}}^{\frac{l}{2}} f(x) \sin \frac{2\pi nx}{l} dx, \quad n \geq 1. \quad (9)$$



The Dirichlet Conditions

Function f satisfies **Dirichlet Conditions** on $\langle a, b \rangle$ if

- 1 f is piecewise monotonic (ie. $\langle a, b \rangle$ can be divided into a finite number of subintervals on which f is monotonic);
- 2 f is continuous on $\langle a, b \rangle$ apart from at most a finite number of points, at which it has discontinuities of the first kind;
- 3 at each point $f(x_0) = \frac{1}{2}(\lim_{x \rightarrow x_0^+} f(x) + \lim_{x \rightarrow x_0^-} f(x))$;
- 4 at the ends $f(a) = f(b) = \frac{1}{2}(\lim_{x \rightarrow a^+} f(x) + \lim_{x \rightarrow b^-} f(x))$.

Imprecisely we may say, that if f has a finite number of finite jumps over some bounded interval, then it can be at any point approximated by its Fourier series.

Example 1: Let $f(x)$ be a function of a period 2π such that

$$f(x) = \begin{cases} 1, & -\pi < x < 0 \\ \frac{1}{2}, & x = -\pi, x = 0, x = \pi \\ 0, & 0 < x < \pi, \end{cases}$$

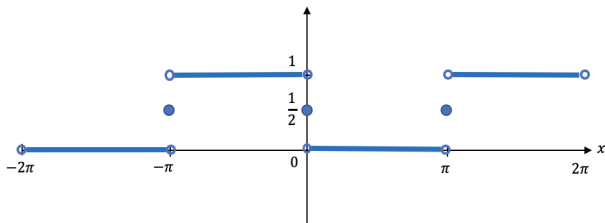
Sketch a graph of f in the interval $-2\pi < x < 2\pi$ and show that the Fourier series of f in the interval $(-\pi, \pi)$ is

$$\frac{1}{2} - \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x + \dots \right]$$



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Example 1: 2π periodic function (part 2)





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Example 1: 2π periodic function (part 3)

Calculating a_0 .

$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx &= \frac{1}{\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{\pi} \int_0^{\pi} f(x) dx \\&= \frac{1}{\pi} \int_{-\pi}^0 1 dx + \frac{1}{\pi} \int_0^{\pi} 0 dx \\&= \frac{1}{\pi} \int_{-\pi}^0 1 dx \\&= \frac{1}{\pi} [x]_{-\pi}^0 \\&= \frac{1}{\pi} (0 - (-\pi)) \\ \text{ie. } a_0 &= 1\end{aligned}$$



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Example 1: 2π periodic function (part 4)

Calculating a_n .

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx dx \\&= \frac{1}{\pi} \int_{-\pi}^0 1 \cos nx dx + \frac{1}{\pi} \int_0^{\pi} 0 \cos nx dx \\&= \frac{1}{\pi} \int_{-\pi}^0 \cos nx dx \\&= \frac{1}{\pi} \left[\frac{\sin nx}{n} \right]_{-\pi}^0 \\&= \frac{1}{n\pi} (\sin(0) - \sin(-n\pi))\end{aligned}$$

ie. $a_n = 0$, for any $n \geq 1$.



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Example 1: 2π periodic function (part 5)

Calculating b_n .

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) \sin nx dx + \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 1 \sin nx dx + \frac{1}{\pi} \int_0^{\pi} 0 \sin nx dx \\ &= \frac{1}{\pi} \int_{-\pi}^0 \sin nx dx \\ &= \frac{1}{\pi} \left[-\frac{\cos nx}{n} \right]_{-\pi}^0 \\ &= -\frac{1}{n\pi} (\cos(0) - \cos(-n\pi)) \end{aligned}$$

That is,

$$\begin{aligned} b_n &= -\frac{1}{n\pi}(1 - \cos(n\pi)) \\ &= -\frac{1}{n\pi}(1 - (-1)^n) \end{aligned}$$

where,

$$b_n = \begin{cases} 0, & n \text{ even} \\ -\frac{2}{n\pi}, & n \text{ odd} \end{cases}$$

since,

$$(-1)^n = \begin{cases} 1, & n \text{ even} \\ -1, & n \text{ odd} \end{cases}$$



Example 1: 2π periodic function (part 7)
Now we need to fit a_0 , a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

From what we have calculated so far

$$a_0 = 1, a_n = 0, b_n = \begin{cases} 0, & n \text{ even} \\ -\frac{2}{n\pi}, & n \text{ odd} \end{cases}$$

By arranging it into a table of values of b_n we obtain

n	1	2	3	4	5
b_n	$-\frac{2}{\pi}$	0	$-\frac{2}{\pi} \left(\frac{1}{3}\right)$	0	$-\frac{2}{\pi} \left(\frac{1}{5}\right)$

Introducing coefficients from the above table into a general formula we obtain

$$f(x) = \frac{1}{2} - \frac{2}{\pi} \left[\sin x + \frac{1}{3} \sin 3x + \frac{1}{5} \sin 5x \dots \right].$$

Example 2:

Let f be a function of a period 2π such that

$$f(x) = x, \text{ for } x \in (-\pi, \pi)$$

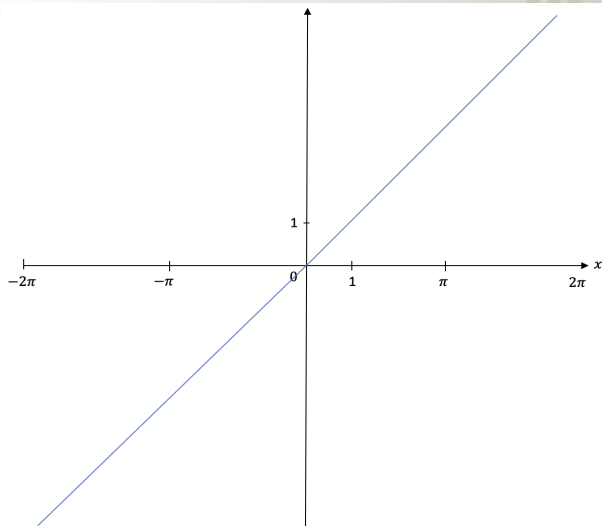
and is extended to a closed interval by putting $f(x) = 0$ at $x = (2k + 1)\pi$, $k \in \mathbb{Z}$. Sketch a graph of f in the interval $-2\pi < x < 2\pi$ and show that the Fourier series of f in the interval $(-\pi, \pi)$ is

$$2 \left[\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right]$$



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Example 2: 2π periodic function (part 2)





AGH Calculating a_0 .

Example 2: 2π periodic odd function (part 3)

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x dx = \frac{1}{2\pi} \left[\frac{x^2}{2} \right]_{-\pi}^{\pi} = 0$$

Calculating a_n , integrating by parts.

We expect a_n to be equal 0, because $x \cos nx$ is an odd function and the integral from the odd function over the symmetric interval cancels out.

$$\begin{aligned} a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \cos nx dx \\ &= \frac{1}{\pi} \left\{ \left[x \frac{\sin nx}{n} \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{\sin nx}{n} dx \right\} \\ &= \frac{1}{\pi} \left\{ \frac{1}{n} (\pi \sin(n\pi) - (-\pi) \sin(-n\pi)) - \frac{1}{n} \int_{-\pi}^{\pi} \sin nx dx \right\} = 0 \end{aligned}$$



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Example 2: 2π periodic odd function (part 4)

Calculating b_n .

$$\begin{aligned} b_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx dx = \frac{1}{\pi} \int_{-\pi}^{\pi} x \sin nx dx \\ &= \frac{1}{\pi} \left\{ \left[-x \frac{\cos nx}{n} \right]_{-\pi}^{\pi} - \int_{-\pi}^{\pi} \frac{-\cos nx}{n} dx \right\} \\ &= \frac{1}{\pi} \left\{ -\frac{1}{n} (\pi \cos(n\pi) - (-\pi) \cos(-n\pi)) + \frac{1}{n} \int_{-\pi}^{\pi} \cos nx dx \right\} \\ &= \frac{1}{\pi} \left\{ -\frac{\pi}{n} (\cos(n\pi) + \cos(n\pi)) + \frac{1}{n} \cdot 0 \right\} \\ &= -\frac{2}{n} \cos n\pi = -\frac{2}{n} (-1)^n \end{aligned}$$



Example 2: 2π periodic odd function (part 5)

Now we need to fit a_0 , a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos nx + b_n \sin nx].$$

From what we have calculated so far (and b_n in the table)

$$a_0 = 0, \quad a_n = 0, \quad b_n = -\frac{2}{n}(-1)^n$$

n	1	2	3
b_n	2	-1	$\frac{2}{3}$

Introducing coefficients from the above table into a general formula we obtain the required series expansion of our function

$$f(x) = 2 \left[\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right].$$

Example 3:

Let f be a function of a period 2π such that

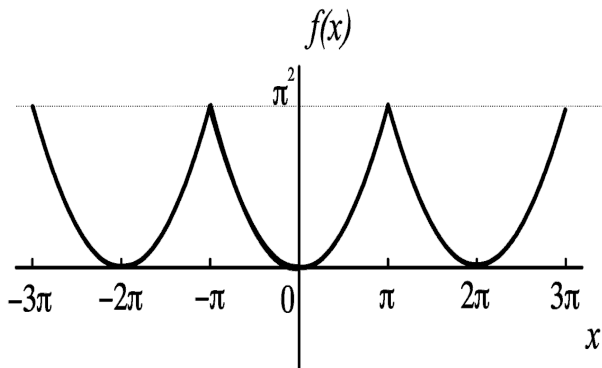
$$f(x) = x^2, \text{ for } x \in (-\pi, \pi)$$

Sketch a graph of f in the interval $-3\pi < x < 3\pi$ and calculate the Fourier series.



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Example 3: 2π periodic even function (part 2)



Calculating a_0 .

$$a_0 = \frac{1}{2\pi} \int_{-\pi}^{\pi} x^2 dx$$

$$\begin{aligned} \int_{-\pi}^{\pi} x^2 dx &= \left[\frac{1}{3} x^3 \right]_{-\pi}^{\pi} \\ &= \frac{(\pi)^3}{3} - \left(\frac{(-\pi)^3}{3} \right) = \frac{2\pi^3}{3} \end{aligned}$$



Example 3: 2π periodic even function (part 4)

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Calculating a_n , integrating by parts.

$$a_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \cos\left(\frac{n\pi x}{\pi}\right) dx$$

$$\int x^2 \cos nx dx = \left| \begin{array}{l} u = nx \\ du = ndx \end{array} \right| = \int x^2 \cos(u) \frac{1}{n} du = \int \frac{x^2 \cos(u)}{n} du \mapsto u = nx \Rightarrow x =$$

$$\frac{u}{n} \mapsto \int \frac{\left(\frac{u}{n}\right)^2 \cos(u)}{n} du = \int \frac{u^2 \cos(u)}{n^3} du = \frac{1}{n^3} \int u^2 \cos(u) du =$$

$$\left| \begin{array}{ll} u = u^2 & v' = \cos(u) \\ u' = 2u & v = \sin(u) \end{array} \right| = \frac{1}{n^3} (u^2 \sin(u) - \int 2u \sin(u) du) = \int 2u \sin(u) du =$$

$$2 \int u \sin(u) du = \left| \begin{array}{ll} u = u & v' = \sin(u) \\ u' = 1 & v = -\cos(u) \end{array} \right| = 2(-u \cos(u) - \int (-\cos(u) du)) =$$

$$2(-u \cos(u) + \sin(u)) = \frac{1}{n^3} (u^2 \sin(u) - 2(-u \cos(u) + \sin(u))) \mapsto u = nx \mapsto$$

$$\frac{1}{n^3} ((nx)^2 \sin(nx) - 2(-nxcos(nx) + \sin(nx))) = \left[\frac{1}{n^3} (n^2 x^2 \sin(nx) -$$

$$2(-nxcos(nx) + \sin(nx)) \right]_{-\pi}^{\pi} = \frac{2\pi(-1)^n}{n^2} - \left(-\frac{2\pi(-1)^n}{n^2}\right) = \frac{4\pi(-1)^n}{n^2}$$



Example 3: 2π periodic even function (part 5)

Calculating b_n . $b_n = \frac{1}{\pi} \int_{-\pi}^{\pi} x^2 \sin\left(\frac{n\pi x}{\pi}\right) dx$

$$\begin{aligned}
 \int x^2 \sin(nx) dx &= \left| \begin{array}{l} u = x^2 \quad v' = \sin(nx) \\ u' = 2x \quad v = -\frac{1}{n} \cos(nx) \end{array} \right| = x^2 \left(-\frac{1}{n} \cos(nx)\right) - \\
 \int 2x \left(-\frac{1}{n} \cos(nx)\right) dx &= -\frac{1}{n} x^2 (\cos(nx)) - \int \left(-\frac{2}{n} x \cos(nx)\right) dx = \\
 \int \left(-\frac{2}{n} x \cos(nx)\right) dx &= -\frac{2}{n} \int x \cos(nx) dx = \left| \begin{array}{l} u = nx \\ du = n dx \end{array} \right| = \int \frac{x \cos(u)}{n} du \mapsto \\
 u = nx \Rightarrow x = \frac{u}{n} \mapsto \int \frac{\frac{u}{n} \cos(u)}{n} du &= -\frac{2}{n} \int \frac{u \cos(u)}{n^2} du = \\
 -\frac{2}{n} \cdot \frac{1}{n^2} \int u \cos(u) du &= \int u \cos(u) du = \left| \begin{array}{l} u = u \quad v' = \cos(u) \\ u' = 1 \quad v = \sin(u) \end{array} \right| = \\
 u \sin(u) - \int \sin(u) du &= -\frac{2}{n} \cdot \frac{1}{n^2} (u \sin(u) - (-\cos(u))) \mapsto u = nx \mapsto \\
 -\frac{2}{n} \cdot \frac{1}{n^2} (nx \sin(nx) - (-\cos(nx))) &= -\frac{2}{n^3} (nx \sin(nx) + \cos(nx)) = \\
 -\frac{1}{n} x^2 (\cos(nx)) - \left(-\frac{2}{n^3} (nx \sin(nx) + \cos(nx))\right) &= \\
 \left[-\frac{1}{n} x^2 (\cos(nx)) + \frac{2}{n^3} (nx \sin(nx) + \cos(nx)) \right]_{-\pi}^{\pi} &=
 \end{aligned}$$



Example 3: 2π periodic even function (part 6)

Now we need to fit a_0, a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

From what we have calculated so far

$$a_0 = \frac{2\pi^3}{3}, \quad a_n = \frac{4\pi(-1)^n}{n^2}, \quad b_n = 0$$

Introducing coefficients from the above, into a general formula we obtain the required series expansion of our function

$$f(x) = \frac{\pi^2}{3} + \sum_{n=1}^{\infty} \frac{4(-1)^n \cos(nx)}{n^2}$$

Example 4:

Let f be a function of a period 2π such that

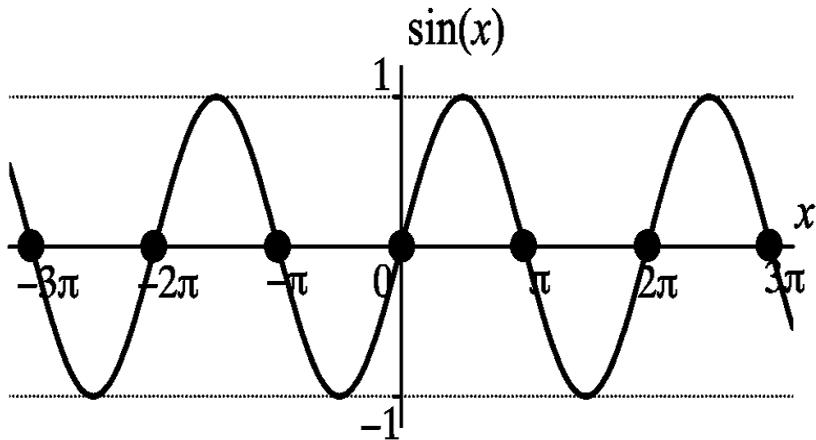
$$f(x) = \sin(ax), \text{ for } x \in (-\pi, \pi), a \in \mathbb{R} \setminus \{\mathbb{Z}\}$$

Sketch a graph of f in the interval $-3\pi < x < 3\pi$ and calculate the Fourier series.



Example 4: 2π periodic odd function (part 2)

A ...



Calculating a_0 .

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_{-\pi}^{\pi} f(x) dx = \\ &= \frac{2}{\pi} \left[\left(-\frac{\cos(a\pi)}{a} + \frac{\cos(-a\pi)}{a} \right) \right]_{-\pi}^{\pi} = \\ &= \frac{2}{\pi} \frac{\cos(-a\pi) - \cos(a\pi)}{a} \end{aligned}$$

Calculating a_n , integrating by parts.

$$\begin{aligned}
 a_n &= \frac{2}{\pi} \int_{-\pi}^{\pi} (f(x) \cos(nx)) dx = \\
 &= \frac{2}{\pi} \int_{-\pi}^{\pi} f(x) \cos(nx) dx = \\
 &= \frac{2}{\pi} \left[\frac{n \sin(ax) \sin(nx) + a \cos(ax) \cos(nx)}{a^2 - n^2} \right]_{-\pi}^{\pi} \\
 &= \\
 &= \frac{2}{\pi} \left(\frac{n \sin(a\pi) \sin(n\pi) + a \cos(a\pi) \cos(n\pi)}{a^2 - n^2} - \frac{n \sin(-a\pi) \sin(-n\pi) + a \cos(-a\pi) \cos(-n\pi)}{a^2 - n^2} \right) = \\
 &= \frac{2}{\pi} \left(\frac{a \cos(a\pi) \cos(n\pi) - \cos(-a\pi) \cos(-n\pi)}{a^2 - n^2} \right) = 0
 \end{aligned}$$

Calculating b_n .

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_{-\pi}^{\pi} f(x) \sin(nx) dx = \\
 &= \frac{2}{\pi} \left[\frac{n \sin(ax) \cos(nx) - a \cos(ax) \sin(nx)}{a^2 - n^2} \right]_{-\pi}^{\pi} \\
 &= \\
 &= \frac{2}{\pi} \left(\frac{n \sin(a\pi) \cos(n\pi) - a \cos(a\pi) \sin(n\pi)}{a^2 - n^2} - \frac{n \sin(-a\pi) \cos(-n\pi) - a \cos(-a\pi) \sin(-n\pi)}{a^2 - n^2} \right) = \\
 &= \frac{2}{\pi} \frac{n \sin(a\pi) \cos(n\pi) - n \sin(-a\pi) \cos(-n\pi)}{a^2 - n^2} \\
 b_n &= \begin{cases} \frac{2n}{\pi} \left[\frac{\sin(a\pi) - \sin(-a\pi)}{a^2 - n^2} \right], & \text{for even } n \\ \frac{2n}{\pi} \left[\frac{\sin(-a\pi) - \sin(a\pi)}{a^2 - n^2} \right], & \text{for odd } n \end{cases}
 \end{aligned}$$

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Example 4: 2π periodic odd function (part 6)

Now we need to fit a_0 , a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

From what we have calculated so far

$$a_0 = \frac{2}{\pi} \frac{\cos(-a\pi) - \cos(a\pi)}{a}$$

$$a_n = 0$$

$$b_n = \frac{2n}{\pi} \left[\frac{\sin(a\pi) - \sin(-a\pi)}{a^2 - n^2} \right] (-1)^n$$

Introducing coefficients from the above, into a general formula we obtain the required series expansion of our function

$$f(x) = \frac{\cos(-a\pi) - \cos(a\pi)}{a\pi} + \sum_{n=1}^{\infty} \frac{2}{\pi} \left[\frac{\sin(a\pi) - \sin(-a\pi)}{a^2 - n^2} \right] (-1)^n \sin(nx)$$

Example 5:

Let f be a function, such that

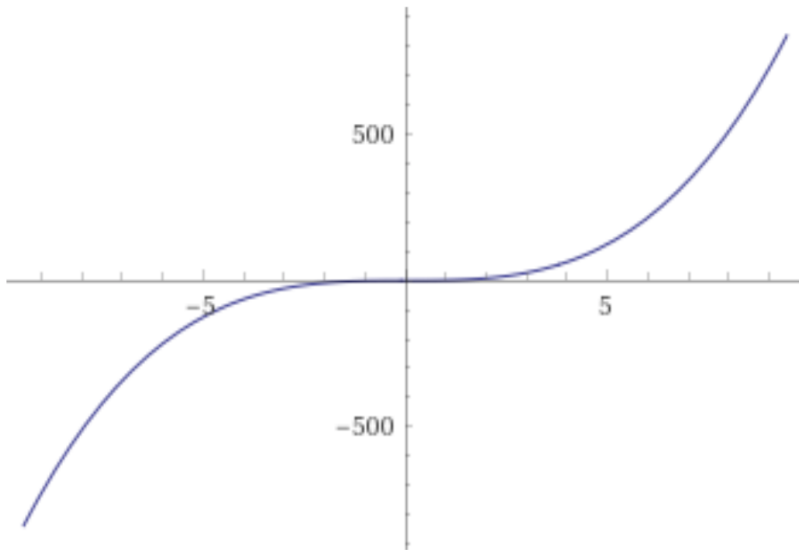
$$f(x) = x^3, \text{ for } x \in (-\pi, \pi)$$

Sketch a graph of f in the interval $-3\pi < x < 3\pi$ and calculate the Fourier series.



Example 5: non periodic odd function (part 2)

A



Calculating a_0 .

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_{-\pi}^{\pi} x^3 dx = \\ &= \frac{2}{\pi} \left[\frac{x^4}{4} \right]_{-\pi}^{\pi} \\ &= \frac{2}{\pi} \left[\frac{\pi^4}{4} - \frac{\pi^4}{4} \right] = 0 \end{aligned}$$



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Example 5: non periodic odd function (part 4)

Calculating a_n , integrating by parts.

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_{-\pi}^{\pi} x^3 \cos\left(\frac{\pi n x}{\pi}\right) dx = \\ &= \frac{2}{\pi} \int_{-\pi}^{\pi} x^3 \cos(nx) dx = \\ &= \frac{2}{\pi} \left[\frac{nx(n^2x^2-6) \sin(nx) + 3(n^2x^2-2) \cos(nx)}{n^4} \right]_{-\pi}^{\pi} \\ &= \frac{2}{\pi} \left[\frac{n\pi(n^2\pi^2-6) \sin(\pi n) + 3(\pi^2n^2-2) \cos(\pi n)}{n^4} \right] - \\ &\quad \frac{2}{\pi} \left[\frac{-\pi n(\pi^2n^2-6) \sin(-\pi n) + 3(\pi^2n^2-2) \cos(-\pi n)}{n^4} \right] = \\ &= \\ &\quad \frac{2}{\pi} \left[\frac{n\pi(n^2\pi^2-6) \sin(\pi n) + 3(\pi^2n^2-2) \cos(\pi n) + \pi n(\pi^2n^2-6) \sin(-\pi n) - 3(\pi^2n^2-2) \cos(-\pi n)}{n^4} \right] = \\ &= \frac{2}{\pi} \left[\frac{n\pi(n^2\pi^2-6) \sin(\pi n) + \pi n(\pi^2n^2-6) \sin(-\pi n)}{n^4} \right] = \\ &= \frac{2}{\pi} \left[\frac{\pi(n^2\pi^2-6) \sin(\pi n) + \pi(\pi^2n^2-6) \sin(-\pi n)}{n^3} \right] = 0 \end{aligned}$$



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Example 5: non periodic odd function (part 5)

Calculating b_n .

$$\begin{aligned} b_n &= \frac{2}{\pi} \int_{-\pi}^{\pi} x^3 \sin\left(\frac{n\pi x}{\pi}\right) dx = \\ &= \frac{2}{\pi} \int_{-\pi}^{\pi} x^3 \sin(nx) dx = \\ &= \frac{2}{\pi} \left[\frac{3(n^2 x^2 - 2) \sin(nx) - nx(n^2 x^2 - 6) \cos(nx)}{n^4} \right]_{-\pi}^{\pi} \\ &= \frac{2}{\pi} \left[\frac{3(n^2 \pi^2 - 2) \sin(n\pi) - n\pi(n^2 \pi^2 - 6) \cos(n\pi)}{n^4} \right] - \\ &\frac{2}{\pi} \left[\frac{3(n^2 \pi^2 - 2) \sin(-\pi n) + \pi n(n^2 \pi^2 - 6) \cos(-\pi n)}{n^4} \right] = \\ &= \frac{2}{\pi} \left[\frac{-n\pi(n^2 \pi^2 - 6) \cos(n\pi) - n\pi(n^2 \pi^2 - 6) \cos(-n\pi)}{n^4} \right] = \\ &= \frac{2}{\pi} \left[-\pi n(n^2 \pi^2 - 6) \right] \left[\frac{\cos(\pi n) + \cos(-\pi n)}{n^4} \right] = \\ &= -2(n^2 \pi^2 - 6) \left[\frac{2 \cos(\pi n)}{n^3} \right] \end{aligned}$$

Calculating b_n , continuation.

$$b_n = \begin{cases} \frac{-4(n^2\pi^2-6)}{n^3}, & \text{for even } n \\ \frac{4(n^2\pi^2-6)}{n^3}, & \text{for odd } n \end{cases}$$



Example 5: 2π non periodic odd function (part 7)

Now we need to fit a_0, a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

From what we have calculated so far

$$a_0 = 0,$$

$$a_n = 0,$$

$$b_n = \frac{-4(n^2\pi^2 - 6)}{n^3}(-1)^n,$$

Introducing coefficients from the above, into a general formula we obtain the required series expansion of our function

$$f(x) = \sum_{n=1}^{\infty} \frac{-4(n^2\pi^2 - 6)}{n^3}(-1)^n \sin(nx)$$

Example 6:

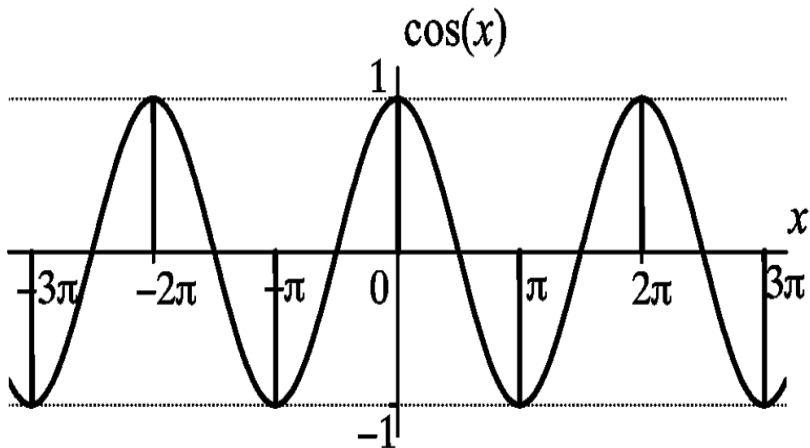
Let f be a function, such that

$$f(x) = \cos(ax), \text{ for } x \in (-\pi, \pi), a \in \mathbb{R} \setminus \{\mathbb{Z}\}$$

Sketch a graph of f in the interval $-3\pi < x < 3\pi$ and calculate the Fourier series.



Example 6: 2π periodic even function (part 2)



Calculating a_0 .

$$\begin{aligned} a_0 &= \frac{2}{\pi} \int_{-\pi}^{\pi} \cos(ax) dx = \\ &= \frac{2}{\pi} \left[\frac{\sin(ax)}{a} \right]_{-\pi}^{\pi} \\ &= \frac{2}{\pi} \frac{\sin(a\pi)}{a} - \frac{2}{\pi} \frac{\sin(-a\pi)}{a} = \\ &= \frac{2}{\pi} \left[\frac{\sin(a\pi) - \sin(-\pi a)}{a} \right] \end{aligned}$$

**Calculating a_n , integrating by parts.**

$$\begin{aligned} a_n &= \frac{2}{\pi} \int_{-\pi}^{\pi} \cos(ax) \cos(nx) dx = \\ &= \frac{2}{\pi} \left[\frac{a \sin(ax) \cos(nx) - n \cos(ax) \sin(nx)}{a^2 - n^2} \right]_{-\pi}^{\pi} \\ &= \frac{2}{\pi} \left[\frac{a \sin(a\pi) \cos(n\pi) - n \cos(a\pi) \sin(n\pi)}{a^2 - n^2} \right] - \\ &\quad \frac{2}{\pi} \left[\frac{a \sin(-a\pi) \cos(-n\pi) - n \cos(-a\pi) \sin(-n\pi)}{a^2 - n^2} \right] = \\ &= \frac{2}{\pi} \left[\frac{a \sin(a\pi) \cos(n\pi) - a \sin(-a\pi) \cos(-n\pi)}{a^2 - n^2} \right] \\ a_n &= \begin{cases} \frac{2}{\pi} \left(\frac{a \sin(a\pi) - a \sin(-a\pi)}{a^2 - n^2} \right), & \text{for even } n \\ \frac{2}{\pi} \left(\frac{a \sin(-a\pi) - a \sin(a\pi)}{a^2 - n^2} \right), & \text{for odd } n \end{cases} \end{aligned}$$

Calculating b_n .

$$\begin{aligned}
 b_n &= \frac{2}{\pi} \int_{-\pi}^{\pi} \cos(ax) \sin(nx) dx = \\
 &= \frac{2}{\pi} \left[\frac{a \sin(ax) \sin(nx) + n \cos(ax) \cos(nx)}{a^2 - n^2} \right]_{-\pi}^{\pi} \\
 &= \frac{2}{\pi} \left[\frac{a \sin(a\pi) \sin(n\pi) + n \cos(a\pi) \cos(n\pi)}{a^2 - n^2} \right] - \\
 &\quad \frac{2}{\pi} \left[\frac{a \sin(-a\pi) \sin(-n\pi) + n \cos(-a\pi) \cos(-n\pi)}{a^2 - n^2} \right] = \\
 &= \frac{2}{\pi} \left[\frac{n \cos(a\pi) \cos(n\pi) - n \cos(-a\pi) \cos(-n\pi)}{a^2 - n^2} \right] = 0
 \end{aligned}$$



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Example 6: 2π periodic even function (part 6)

Now we need to fit a_0 , a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

From what we have calculated so far: $a_0 = \frac{2}{\pi} \left[\frac{\sin(a\pi) - \sin(-\pi a)}{a} \right]$,

$$a_n = \frac{2}{\pi} \left(\frac{a \sin(a\pi) - a \sin(-a\pi)}{a^2 - n^2} \right) (-1)^n,$$

$$b_n = 0,$$

Introducing coefficients from the above, into a general formula we obtain the required series expansion of our function

$$f(x) = \frac{a \sin(a\pi) - a \sin(-a\pi)}{a\pi} + \sum_{n=1}^{\infty} \frac{2}{\pi} \left[\frac{a \sin(a\pi) - a \sin(-a\pi)}{a^2 - n^2} \right] (-1)^n \cos(nx).$$

Example 7:

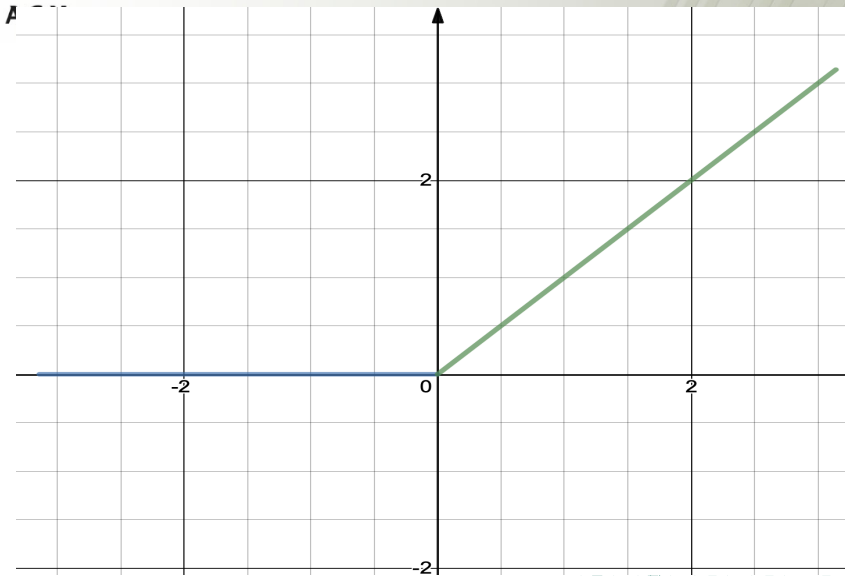
Let f be a function, such that

$$f(x) = \begin{cases} 0, & x \in \langle -\pi; 0 \rangle \\ x, & x \in \langle 0; \pi \rangle \end{cases}$$

Sketch a graph of f in the interval $-\pi < x < \pi$ and calculate the Fourier series.



Example 7: non periodic function (part 2)





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Example 7: non periodic function (part 3)

Calculating a_0 .

$$\begin{aligned}a_0 &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) dx + \frac{1}{\pi} \int_0^{\pi} f(x) dx \\&= \frac{1}{\pi} \int_{-\pi}^0 0 \cdot dx + \frac{1}{\pi} \int_0^{\pi} x dx \\&= \frac{1}{\pi} \left[\frac{x^2}{2} \right]_0^{\pi} = \\&= \frac{1}{\pi} \left(\frac{\pi^2}{2} - 0 \right) = \frac{\pi}{2}.\end{aligned}$$



Example 7: non periodic function (part 4)

Calculating a_n , integrating by parts.

$$\begin{aligned}a_n &= \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \cos nx \, dx \\&= \frac{1}{\pi} \int_{-\pi}^0 f(x) \cos nx \, dx + \frac{1}{\pi} \int_0^{\pi} f(x) \cos nx \, dx \\&= \frac{1}{\pi} \int_{-\pi}^0 0 \cdot \cos nx \, dx + \frac{1}{\pi} \int_0^{\pi} x \cos nx \, dx \\&= \frac{1}{\pi} \int_0^{\pi} x \cos nx \, dx = \frac{1}{\pi} \left\{ \left[x \frac{\sin nx}{n} \right]_0^{\pi} - \int_0^{\pi} \frac{\sin nx}{n} dx \right\} \\&= \frac{1}{\pi} \left\{ \left(\pi \frac{\sin n\pi}{n} - 0 \right) - \frac{1}{n} \left[-\frac{\cos nx}{n} \right]_0^{\pi} \right\} \\&= \frac{1}{\pi} \left\{ (0 - 0) + \frac{1}{n^2} [\cos nx]_0^{\pi} \right\} \\&= \frac{1}{\pi n^2} \{ \cos n\pi - \cos 0 \} = \frac{1}{\pi n^2} \{ (-1)^n - 1 \}\end{aligned}$$

Calculating a_n , continuation.

$$a_n = \begin{cases} 0, & \text{for even } n \\ -\frac{2}{\pi n^2}, & \text{for odd } n \end{cases}$$

Calculating b_n .

$$b_n \stackrel{\text{AGH}}{=} \frac{1}{\pi} \int_{-\pi}^{\pi} f(x) \sin nx \, dx = \frac{1}{\pi} \int_{-\pi}^0 f(x) \sin nx \, dx + \frac{1}{\pi} \int_0^{\pi} f(x) \sin nx \, dx$$

$$= \frac{1}{\pi} \int_{-\pi}^0 0 \cdot \sin nx \, dx + \frac{1}{\pi} \int_0^{\pi} x \sin nx \, dx$$

$$= \frac{1}{\pi} \int_0^{\pi} x \sin nx \, dx = \frac{1}{\pi} \left\{ \left[x \left(-\frac{\cos nx}{n} \right) \right]_0^{\pi} - \int_0^{\pi} \left(-\frac{\cos nx}{n} \right) dx \right\}$$

$$b_n = \frac{1}{\pi} \left\{ -\frac{1}{n} [x \cos nx]_0^{\pi} + \frac{1}{n} \int_0^{\pi} \cos nx \, dx \right\}$$

$$= \frac{1}{\pi} \left\{ -\frac{1}{n} (\pi \cos n\pi - 0) + \frac{1}{n} \left[\frac{\sin nx}{n} \right]_0^{\pi} \right\}$$

$$= -\frac{1}{n} (-1)^n + \frac{1}{\pi n^2} (0 - 0) = -\frac{1}{n} (-1)^n$$

Calculating b_n , continuation.

$$b_n = \begin{cases} -\frac{1}{n}, & \text{for even } n \\ \frac{1}{n}, & \text{for odd } n \end{cases}$$



Example 7: non periodic function (part 8)

Now we need to fit a_0, a_n and b_n into the formula

$$f(x) = \frac{a_0}{2} + \sum_{n=1}^{\infty} [a_n \cos(nx) + b_n \sin(nx)].$$

From what we have calculated so far

$$a_0 = \frac{\pi}{2}, a_n = \frac{1}{\pi n^2} \{\cos n\pi - \cos 0\} = \frac{1}{\pi n^2} \{(-1)^n - 1\}, b_n = -\frac{1}{n} (-1)^n$$

Introducing coefficients from the above, into a general formula we obtain the required series expansion of our function where:

$$a_n = \begin{cases} 0, & \text{for even } n \\ -\frac{2}{\pi n^2}, & \text{for odd } n \end{cases}, \quad b_n = \begin{cases} -\frac{1}{n}, & \text{for even } n \\ \frac{1}{n}, & \text{for odd } n \end{cases}$$

So we have:

$$f(x) = \frac{\pi}{4} -$$

$$\frac{2}{\pi} \left[\cos x + \frac{1}{3^2} \cos 3x + \frac{1}{5^2} \cos 5x + \dots \right] + \left[\sin x - \frac{1}{2} \sin 2x + \frac{1}{3} \sin 3x - \dots \right]$$